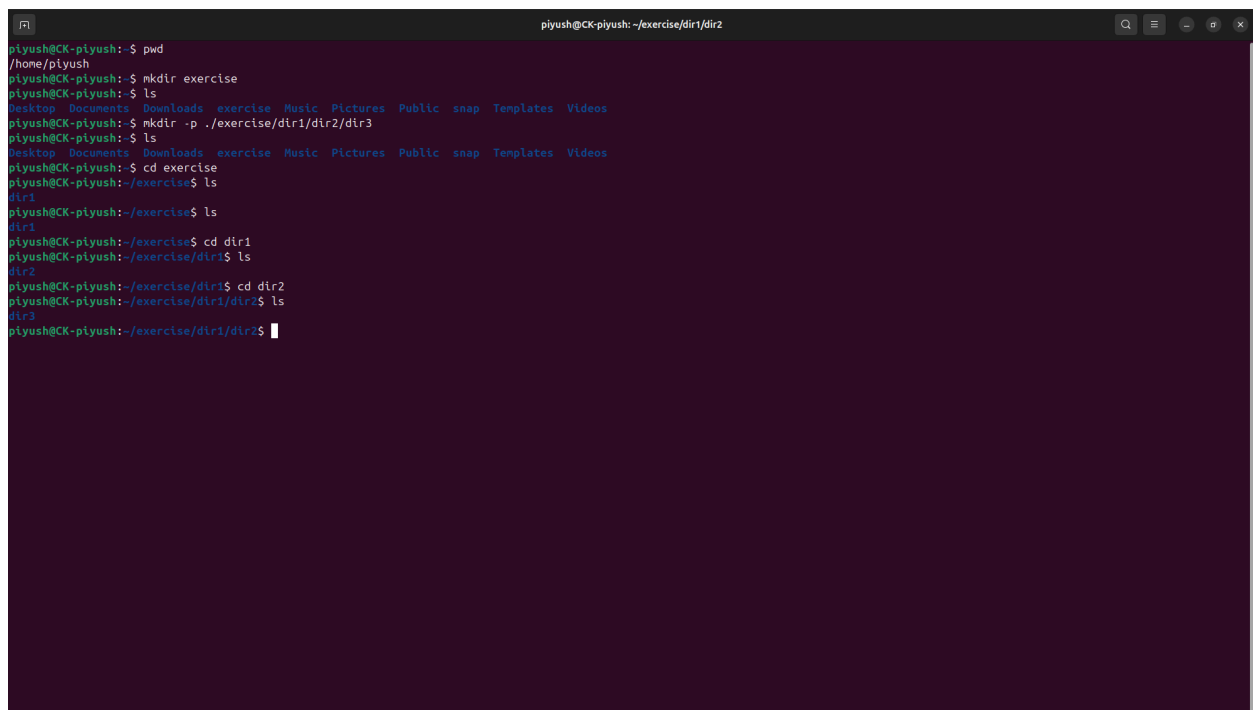


1. Create a directory "exercise" inside your home directory and create nested(dir1/dir2/dir3) directory structure inside "exercise" with single command.

Ans : For the first part of the question I will use the pwd command to check that if I am in the current directory i.e home directory in my case /home/piyush and then I will use the mkdir command to create exercise directory.

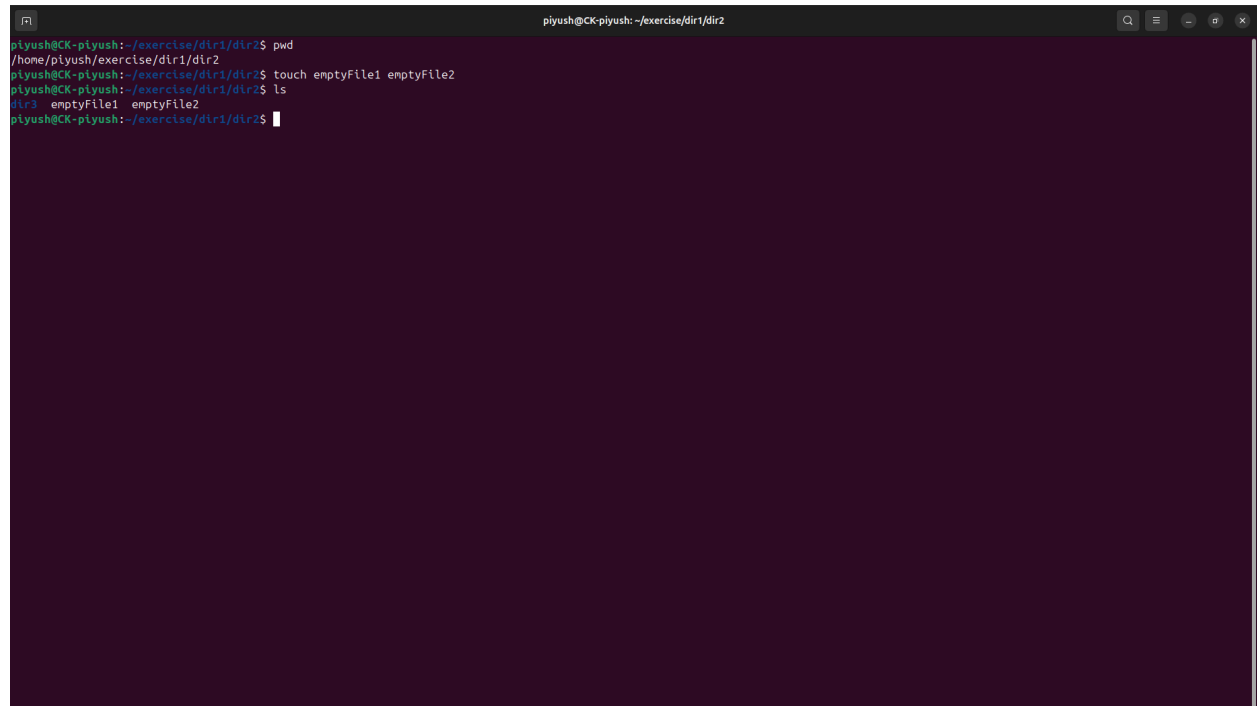
Then I will use mkdir -p to create the dir1/dir2/dir2 directory structure in a single command , because -p option will allow me to create directory if it does not exist.

A terminal window titled 'piyush@CK-piyush: ~/exercise/dir1/dir2' showing a series of commands and their outputs. The user starts in their home directory, creates the 'exercise' directory, and then creates a nested structure 'dir1/dir2/dir3' using the 'mkdir -p' command. Finally, they navigate into each directory level using 'cd' and verify the structure with 'ls'.

```
piyush@CK-piyush:~$ pwd
/home/piyush
piyush@CK-piyush:~$ mkdir exercise
piyush@CK-piyush:~$ ls
Desktop Documents Downloads exercise Music Pictures Public snap Templates Videos
piyush@CK-piyush:~$ mkdir -p ./exercise/dir1/dir2/dir3
piyush@CK-piyush:~$ ls
Desktop Documents Downloads exercise Music Pictures Public snap Templates Videos
piyush@CK-piyush:~$ cd exercise
piyush@CK-piyush:~/exercise$ ls
dir1
piyush@CK-piyush:~/exercise$ ls
dir1
piyush@CK-piyush:~/exercise$ cd dir1
piyush@CK-piyush:~/exercise/dir1$ ls
dir2
piyush@CK-piyush:~/exercise/dir1$ cd dir2
piyush@CK-piyush:~/exercise/dir1/dir2$ ls
dir3
piyush@CK-piyush:~/exercise/dir1/dir2$
```

2. Create two empty files inside dir2 directory: emptyFile1, emptyFile2 in single command .

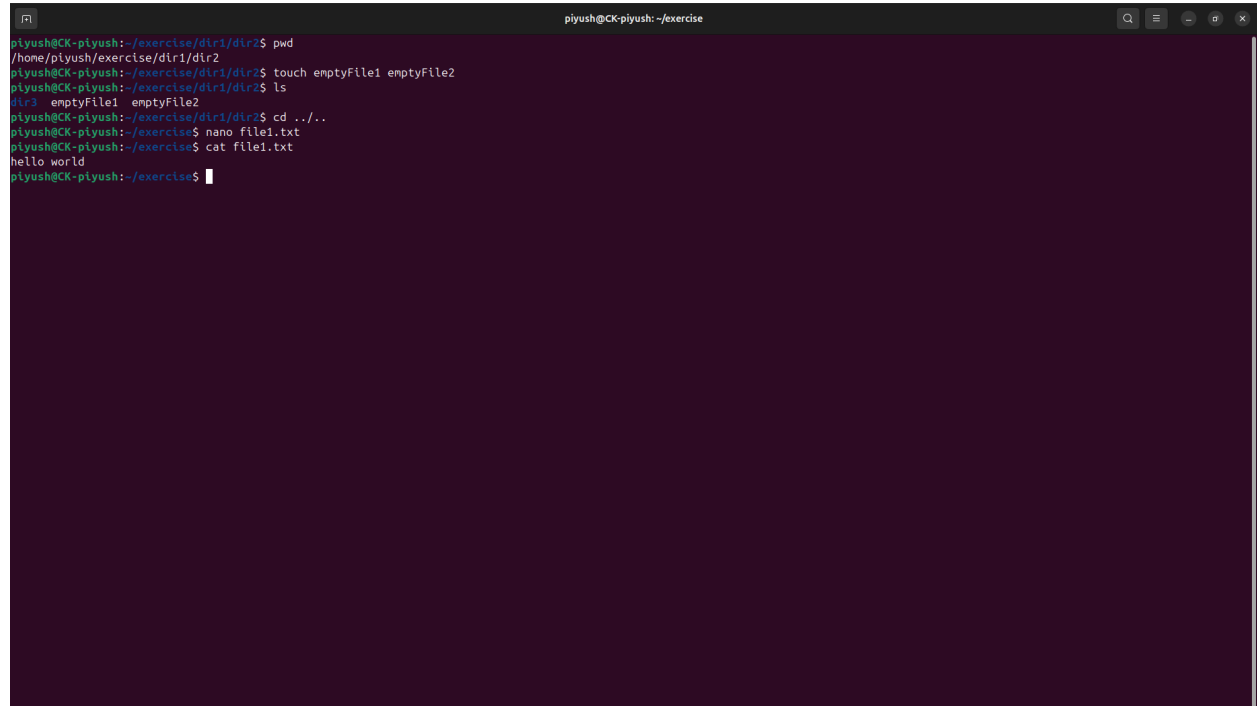
Ans : I will use touch command to create the files , it will let me create the files in a single command .

A terminal window with a dark purple background. The window title is "piyush@CK-piyush: ~/exercise/dir1/dir2". The terminal shows the following commands and output:

```
piyush@CK-piyush: ~/exercise/dir1/dir2$ pwd
/home/piyush/exercise/dir1/dir2
piyush@CK-piyush: ~/exercise/dir1/dir2$ touch emptyFile1 emptyFile2
piyush@CK-piyush: ~/exercise/dir1/dir2$ ls
dir3  emptyFile1  emptyFile2
piyush@CK-piyush: ~/exercise/dir1/dir2$
```

3. Create one file file1.txt containing text "hello world" and save it.

Ans : I will first go back to exercise directory and then use nano command to create the file , editor fill open and then I can write the content in it.

A terminal window with a dark purple background and light green text. The window title is 'piyush@CK-piyush: ~/exercise'. The terminal shows a sequence of commands and their outputs: 'pwd' returns '/home/piyush/exercise/dir1/dir2'; 'touch emptyFile1 emptyFile2' creates two files; 'ls' lists 'dir3', 'emptyFile1', and 'emptyFile2'; 'cd ../../' moves to the parent directory; 'nano file1.txt' opens a nano editor (not visible); 'cat file1.txt' outputs 'hello world'; and the prompt returns to the shell.

```
piyush@CK-piyush: ~/exercise
piyush@CK-piyush:~/exercise/dir1/dir2$ pwd
/home/piyush/exercise/dir1/dir2
piyush@CK-piyush:~/exercise/dir1/dir2$ touch emptyFile1 emptyFile2
piyush@CK-piyush:~/exercise/dir1/dir2$ ls
dir3  emptyFile1  emptyFile2
piyush@CK-piyush:~/exercise/dir1/dir2$ cd ../../
piyush@CK-piyush:~/exercise$ nano file1.txt
piyush@CK-piyush:~/exercise$ cat file1.txt
hello world
piyush@CK-piyush:~/exercise$
```

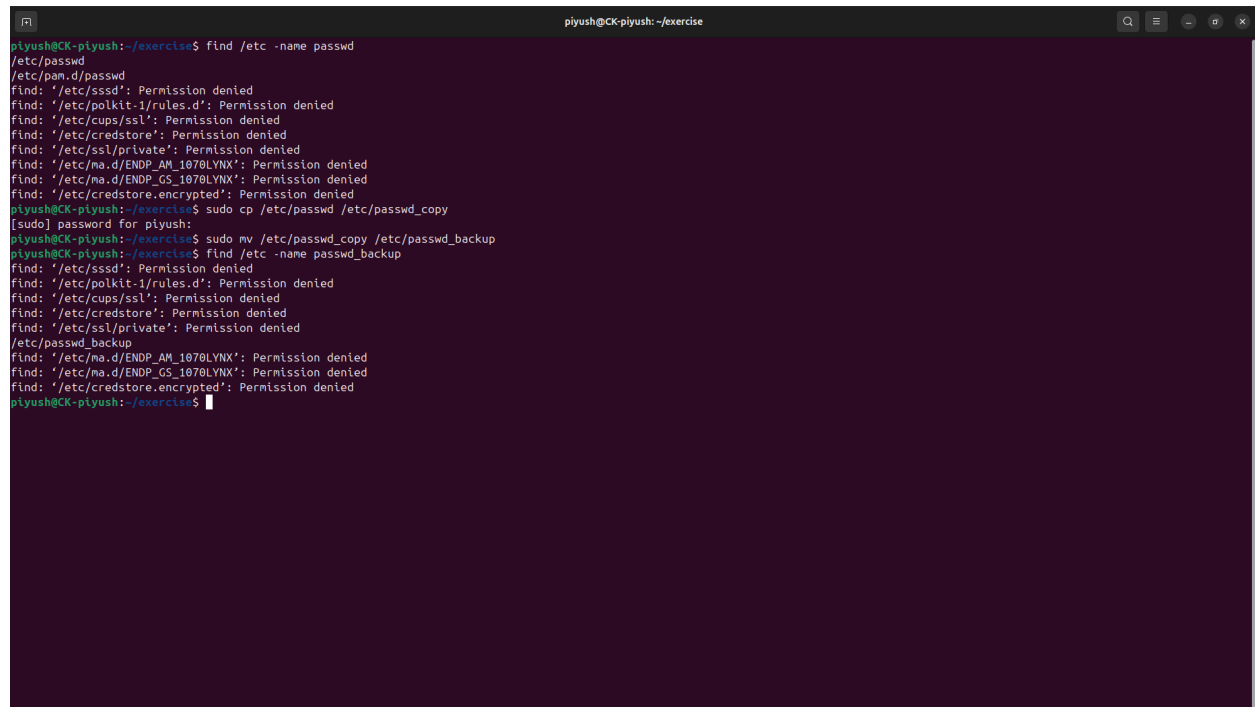
4. Find a "passwd" file using find command inside /etc. copy this files as passwd_copy and then rename this file as passwd_backup.

Ans : For the first part I will use the find command to search for a file in the directory because this command allows user to search for any file in a specific directory.

For the second part I will use cp command to copy the files as passwd_copy.

For the third part of the question I will use mv command to rename the file name to the passwd_backup.

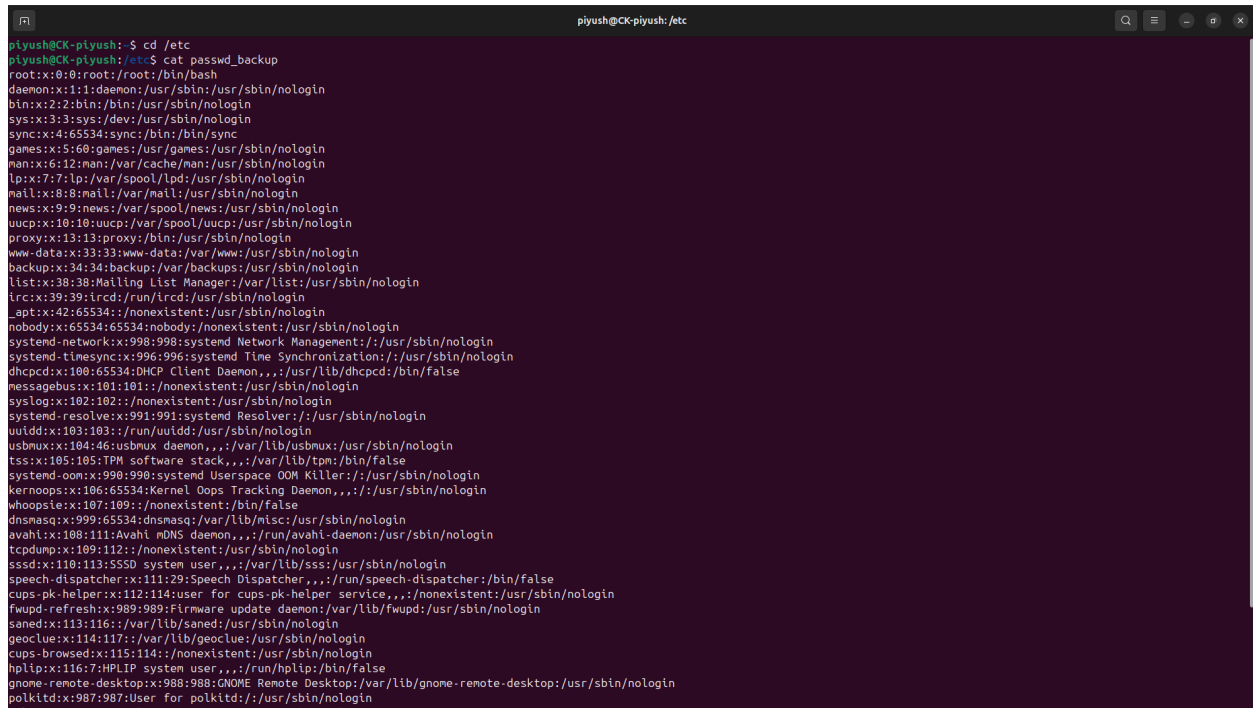
I will have to use sudo command to copy and rename the file as it gives permission denied error if I do it simply cp and mv because the current directory I am in is exercise and the linux does not allow me to do that.



```
piyush@CK-piyush:~/exercise$ find /etc -name passwd
/etc/passwd
/etc/pam.d/passwd
find: '/etc/ssh': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/cups/ssl': Permission denied
find: '/etc/credstore': Permission denied
find: '/etc/ssl/private': Permission denied
find: '/etc/ma.d/ENDP_AM_1070LYNX': Permission denied
find: '/etc/ma.d/ENDP_GS_1070LYNX': Permission denied
find: '/etc/credstore.encrypted': Permission denied
piyush@CK-piyush:~/exercise$ sudo cp /etc/passwd /etc/passwd_copy
[sudo] password for piyush:
piyush@CK-piyush:~/exercise$ sudo mv /etc/passwd_copy /etc/passwd_backup
piyush@CK-piyush:~/exercise$ find /etc -name passwd_backup
find: '/etc/ssh': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/cups/ssl': Permission denied
find: '/etc/credstore': Permission denied
find: '/etc/ssl/private': Permission denied
/etc/passwd_backup
find: '/etc/ma.d/ENDP_AM_1070LYNX': Permission denied
find: '/etc/ma.d/ENDP_GS_1070LYNX': Permission denied
find: '/etc/credstore.encrypted': Permission denied
piyush@CK-piyush:~/exercise$
```

5. Try reading passwd_backup file in multiple tools: less,more,cat,strings etc and find the difference in their usage.

Ans : cat command throws all the file content at once not giving proper navigation and easy to read view , for small files it is good but for large files it may not come as handy.



```
piyush@CK-piyush: /etc$ cat passwd_backup
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-networkd:x:998:998:systemd Network Management:/usr/sbin/nologin
systemd-timesyncd:x:996:996:systemd Time Synchronization:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd/bin/false
messagebus:x:101:101:/nonexistent:/usr/sbin/nologin
syslog:x:102:102:/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/usr/sbin/nologin
uidd:x:103:103:/run/uidd:/usr/sbin/nologin
usbmuxd:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oomd:x:990:990:systemd Userspace OOM Killer:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsie:x:107:109:/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112:/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116:/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117:/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114:/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
```

The less and more command works in a similar way but the less command opens a different screen altogether to view the content of the screen and keeping our terminal clean and not messy . It also gives us with navigation options in a file . The more command opens the file in the terminal itself behaving as cat but giving navigation down option only no up option to view the file.

```
piyush@CK-piyush: /etc$ less /etc/passwd_backup
piyush@CK-piyush: /etc$ more /etc/passwd_backup
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:MailList Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/:nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101:/:nonexistent:/usr/sbin/nologin
syslog:x:102:102:/:nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:usr/sbin/nologin
uiddd:x:103:103:/:run/uiddd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsie:x:107:109:/:nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112:/:nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:20:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116:/:var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117:/:var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114:/:nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
```

The strings command does the same function as cat command it also throws the file content on the terminal .
So amongst all the options I think less command is a better option to view the file.

```
piyush@CK-piyush: /etc$ strings /etc/passwd_backup
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:MailList Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/:nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101:/:nonexistent:/usr/sbin/nologin
syslog:x:102:102:/:nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:usr/sbin/nologin
uiddd:x:103:103:/:run/uiddd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsie:x:107:109:/:nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112:/:nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:20:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116:/:var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117:/:var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114:/:nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
```

6. Find out the number of line in password_backup containing "/bin/false".

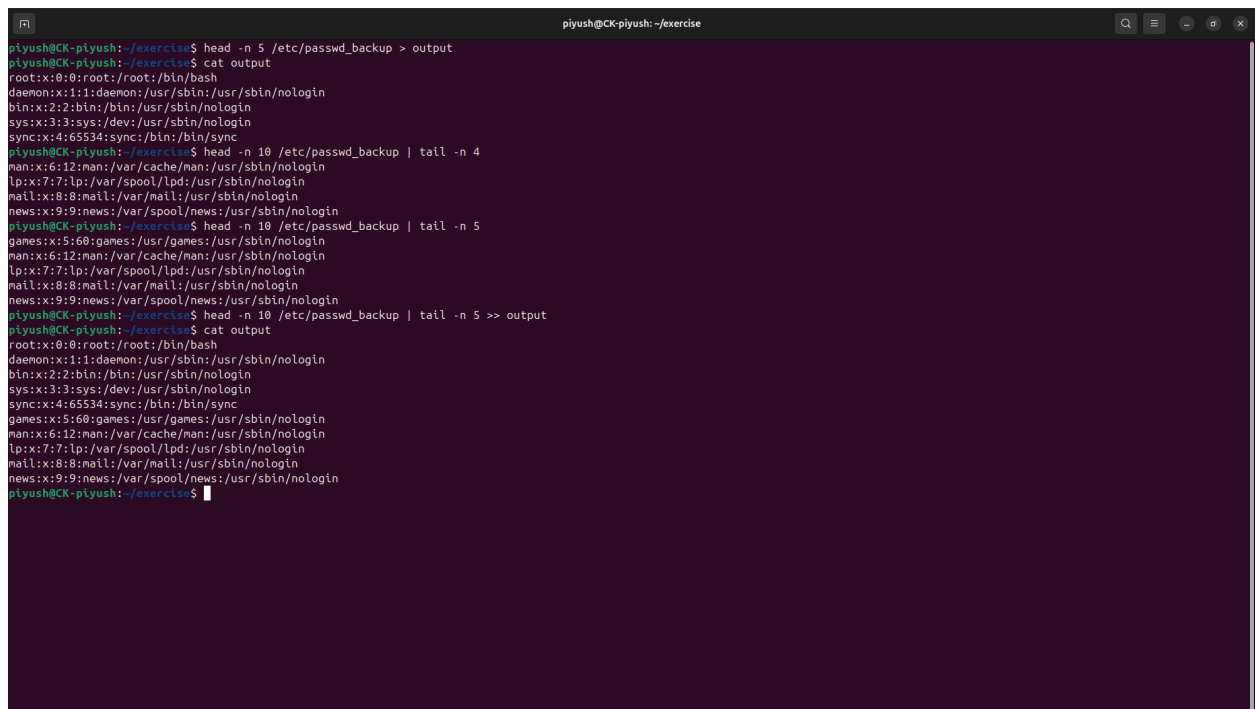
Ans : The command used to count the number of lines in passwd_backup file with occurrence of "/bin/false" is grep used with flag -c to count all the number of occurrences.

```
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:MailList Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/usr/sbin/nologin
uiddd:x:103:103:/run/uiddd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,:/usr/sbin/nologin
whoopsie:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112:/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116:/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117:/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114:/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534:/run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
piyush:x:1000:1000:Piyush:/home/piyush:/bin/bash
nfe:x:997:1001::/home/nfe:/sbin/nologin
piyush@CK-piyush:/etc$ grep -c "/bin/false" passwd_backup
7
piyush@CK-piyush:/etc$
```

7. Get the first 5 lines of a file "password_backup" and Redirect the output of the above commands into file "output". Also, get the lines 6-10 from the above file.

Ans : For the first part of the question , the head command can be used with flag -n and > redirection command can be used to redirect the content to file named output.

For the second part of the question , we can first use the head command with -n 10 then use the pipe command to pipe the output into tail command with -n 5 to get last 6-10 lines and then again use pipe with >> command to redirect the content to output file and append the last 6-10 files.



```
piyush@CK-piyush: ~/exercise
piyush@CK-piyush:~/exercise$ head -n 5 /etc/passwd_backup > output
piyush@CK-piyush:~/exercise$ cat output
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
piyush@CK-piyush:~/exercise$ head -n 10 /etc/passwd_backup | tail -n 4
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
piyush@CK-piyush:~/exercise$ head -n 10 /etc/passwd_backup | tail -n 5
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
piyush@CK-piyush:~/exercise$ head -n 10 /etc/passwd_backup | tail -n 5 >> output
piyush@CK-piyush:~/exercise$ cat output
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
piyush@CK-piyush:~/exercise$
```


8. Create a "test" user, create its password and find out its uid and gid.

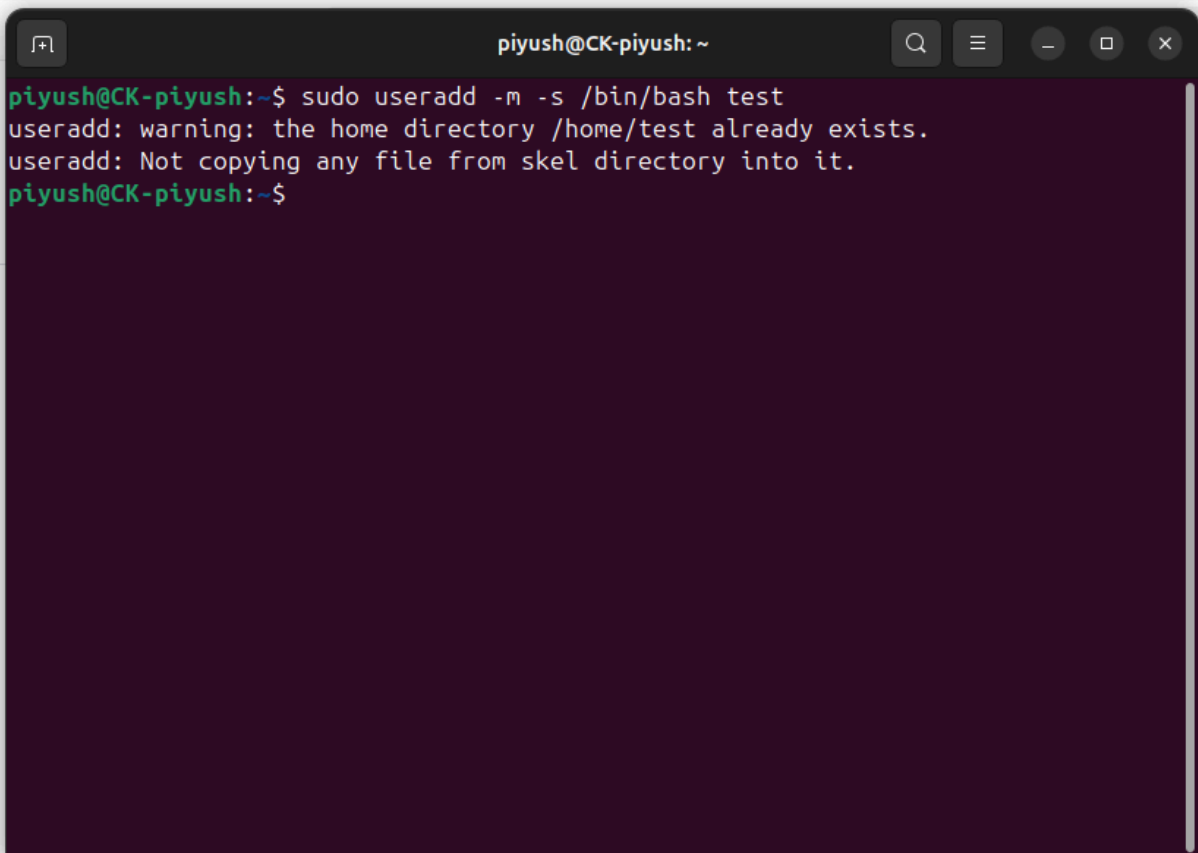
Ans : To create the user : `sudo useradd test` command is used.

To create the user password : `sudo passwd test` command is used,

To find out the uid and gid `cat /etc/passwd` is used which list all the users in the system along with their UID and GID . So We can easily see the UID and GID from the `cat /etc/passwd` command. I can also see the UID and GID by using the Command `id` followed by the username.

The alternate approach to create a new user is to use command `adduser` because by default it will not create home directory . So to create user with configurations , `adduser` command can be used .

The command I used here `sudo useradd -m -s /bin/bash test` : It creates the home directory .

A terminal window titled 'piyush@CK-piyush: ~' with standard window controls. The terminal shows the command 'sudo useradd -m -s /bin/bash test' being executed. The output consists of two lines: 'useradd: warning: the home directory /home/test already exists.' and 'useradd: Not copying any file from skel directory into it.' The prompt returns to 'piyush@CK-piyush:~\$'.

```
piyush@CK-piyush:~$ sudo useradd -m -s /bin/bash test
useradd: warning: the home directory /home/test already exists.
useradd: Not copying any file from skel directory into it.
piyush@CK-piyush:~$
```

Shift +

Select an area to

Saves to the Pictures folder

```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ sudo useradd -m -s /bin/bash test  
useradd: warning: the home directory /home/test already exists.  
useradd: Not copying any file from skel directory into it.  
piyush@CK-piyush:~$ sudo passwd test  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: password updated successfully  
piyush@CK-piyush:~$
```

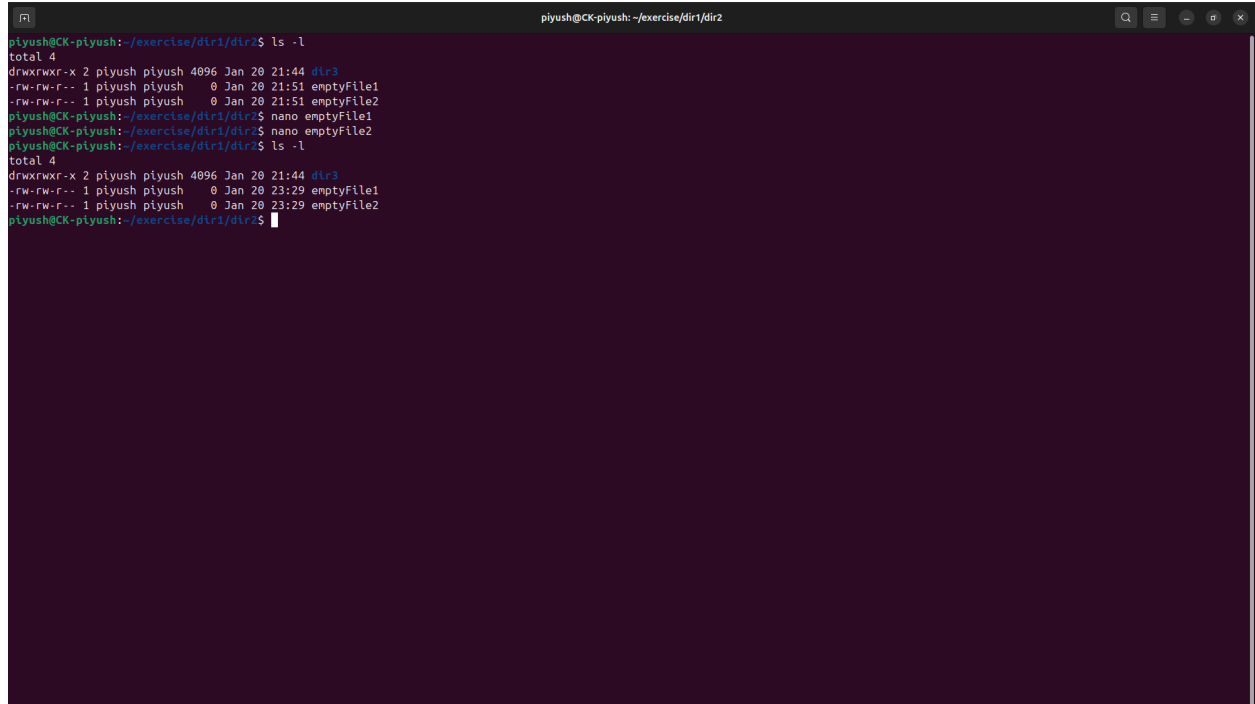
```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ sudo useradd -m -s /bin/bash test  
useradd: warning: the home directory /home/test already exists.  
useradd: Not copying any file from skel directory into it.  
piyush@CK-piyush:~$ sudo passwd test  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: password updated successfully  
piyush@CK-piyush:~$ id test  
uid=1001(test) gid=1002(test) groups=1002(test)  
piyush@CK-piyush:~$
```

9. Change the timestamp of emptyFile1, emptyFile2 which are exist in dir2.

Ans : To change the timestamp I can try to open the file using nano and then just save it without adding any content. This will modify the timestamp simply .

I can also simply use touch command to modify the timestamp in single command .

By doing ' touch dir2/emptyFile1 dir2/emptyFile2 ' , this will also modify the time stamp of the files .



```
piyush@CK-piyush: ~/exercise/dir1/dir2
piyush@CK-piyush:~/exercise/dir1/dir2$ ls -l
total 4
drwxrwxr-x 2 piyush piyush 4096 Jan 20 21:44 dir3
-rw-rw-r-- 1 piyush piyush   0 Jan 20 21:51 emptyFile1
-rw-rw-r-- 1 piyush piyush   0 Jan 20 21:51 emptyFile2
piyush@CK-piyush:~/exercise/dir1/dir2$ nano emptyFile1
piyush@CK-piyush:~/exercise/dir1/dir2$ nano emptyFile2
piyush@CK-piyush:~/exercise/dir1/dir2$ ls -l
total 4
drwxrwxr-x 2 piyush piyush 4096 Jan 20 21:44 dir3
-rw-rw-r-- 1 piyush piyush   0 Jan 20 23:29 emptyFile1
-rw-rw-r-- 1 piyush piyush   0 Jan 20 23:29 emptyFile2
piyush@CK-piyush:~/exercise/dir1/dir2$
```

10. Login as test user and edit the "output" file created above. Since the permission won't allow you to save the changes. Configure such that test user can edit it.

- a. Add group owner of the "output" file as the secondary group of testuser and check/change the "output" file permission if it is editable by group. Once done revert the changes
- b. Make the file editable to the world so that test user can access it. Revert the changes after verification
- c. Change the ownership to edit the file.

Ans : a. The file is owned by home user i.e me in this case and the test user has not any access to edit it. So to add group owner I use

' `sudo usermod -aG piyush test` ' command to add the test user as the secondary group of piyush i.e owner ' . This is all done as current user being piyush.

To check if the group is added I use `groups test` command to see if test has been added .

To allow the group write permission I can use the command `chmod g+rw output` And the permissions are given. Now To revert the changes done I do `sudo gpasswd -d test piyush` to remove test from the group and change the permission back by `chmod g-rw output` .

b. Now to make the file editable to the world simply give rw permission to the others by using command `chmod o+rw output` . To revert the changes back I Use the command `chmod o-rw output`.

c. Now to change the ownership of the file to edit . I will use the command `sudo chown test output` , this will make the test user as the owner of the file. To revert back the changes the command `sudo chown piyush output` will be used.

test: cannot open directory /home/test: Permission denied

piyush@CK-piyush:~\$ su - test

Password:

test@CK-piyush:~\$ ls -la ~

total 20

drwxr-x--- 2 test test 4096 Jan 21 20:02 .

drwxr-xr-x 5 root root 4096 Jan 21 20:02 ..

-rw-r--r-- 1 test test 220 Mar 31 2024 .bash_logout

-rw-r--r-- 1 test test 3771 Mar 31 2024 .bashrc

-rw-r--r-- 1 test test 807 Mar 31 2024 .profile

test@CK-piyush:~\$ sudo usermod -aG piyush test

[sudo] password for test:

test is not in the sudoers file.

test@CK-piyush:~\$ exit

logout

piyush@CK-piyush:~\$ sudo usermod -aG piyush test

piyush@CK-piyush:~\$ groups test

test : test piyush

piyush@CK-piyush:~\$ █

GNU nano 7.2

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
Hello World
```

```
piyush@CK-piyush:~/exercise$ sudo gpasswd -d test piyush
Removing user test from group piyush
piyush@CK-piyush:~/exercise$ chmod g-rw output
piyush@CK-piyush:~/exercise$
```

```
piyush@CK-piyush:~/exercise$ chmod o+rw output
piyush@CK-piyush:~/exercise$ ls -al
total 20
drwxrwxr-x  3 piyush piyush 4096 Jan 21 20:19 .
drwxr-x--- 16 piyush piyush 4096 Jan 21 17:09 ..
drwxrwxr-x  3 piyush piyush 4096 Jan 20 21:44 dir1
-rw-rw-r--  1 piyush piyush  12 Jan 20 21:54 file1.txt
-rw----rw-  1 piyush piyush  436 Jan 21 20:14 output
piyush@CK-piyush:~/exercise$ su - test
Password:
test@CK-piyush:~$ vi /home/piyush/exercise/output
test@CK-piyush:~$ exit
logout
piyush@CK-piyush:~/exercise$ chmod o-rw output
piyush@CK-piyush:~/exercise$ ls -al
total 20
drwxrwxr-x  3 piyush piyush 4096 Jan 21 20:19 .
drwxr-x--- 16 piyush piyush 4096 Jan 21 17:09 ..
drwxrwxr-x  3 piyush piyush 4096 Jan 20 21:44 dir1
-rw-rw-r--  1 piyush piyush  12 Jan 20 21:54 file1.txt
-rw-----  1 piyush piyush  436 Jan 21 20:14 output
piyush@CK-piyush:~/exercise$
```

```
-rw----- 1 piyush piyush 436 Jan 21 20:14 output
piyush@CK-piyush:~/exercise$ sudo chown test output
piyush@CK-piyush:~/exercise$ ls -l output
-rw----- 1 test piyush 436 Jan 21 20:14 output
piyush@CK-piyush:~/exercise$
```

```
piyush@CK-piyush:~/exercise$ su - test
Password:
test@CK-piyush:~$ vi /home/piyush/exercise/output
test@CK-piyush:~$ exit
logout
piyush@CK-piyush:~/exercise$ sudo chown piyush output
piyush@CK-piyush:~/exercise$ ls -al
total 20
drwxrwxr-x  3 piyush piyush 4096 Jan 21 20:19 .
drwxr-x--- 16 piyush piyush 4096 Jan 21 17:09 ..
drwxrwxr-x  3 piyush piyush 4096 Jan 20 21:44 dir1
-rw-rw-r--  1 piyush piyush  12 Jan 20 21:54 file1.txt
-rw-----  1 piyush piyush  436 Jan 21 20:14 output
piyush@CK-piyush:~/exercise$
```

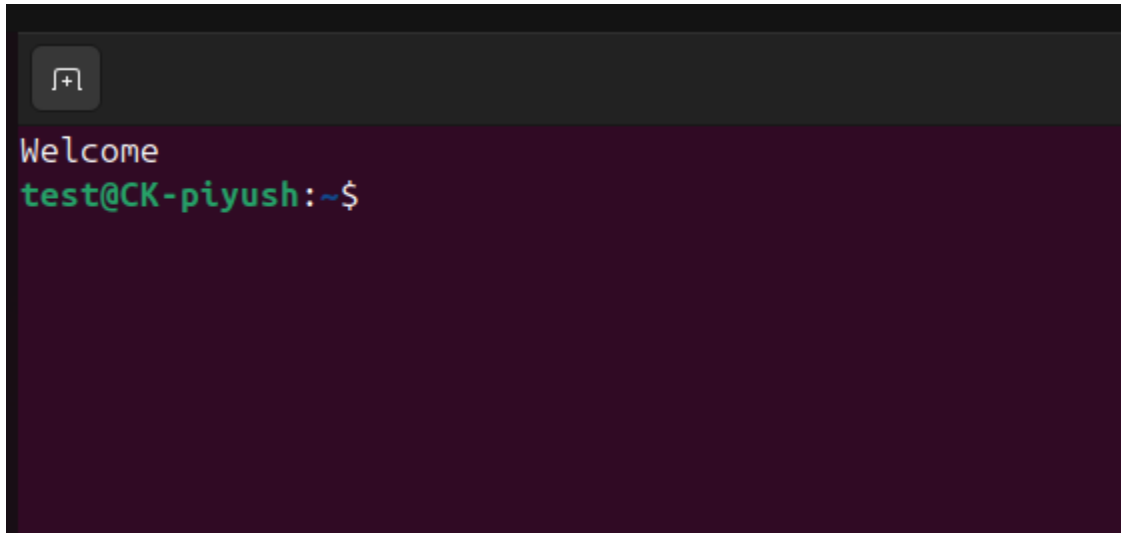

11. Create alias with your name so that it creates a file as "/tmp/aliastesting".
Ans: To do this the alias command can be used .

```
piyush@CK-piyush: ~/exercise
piyush@CK-piyush:~/exercise$ alias piyush="touch /tmp/aliastesting"
piyush@CK-piyush:~/exercise$ ls -l
total 12
drwxrwxr-x 3 piyush piyush 4096 Jan 20 21:44 dir1
-rw-rw-r-- 1 piyush piyush  12 Jan 20 21:54 file1.txt
-rw-rw-r-- 1 piyush piyush 424 Jan 20 23:04 output
piyush@CK-piyush:~/exercise$ piyush
piyush@CK-piyush:~/exercise$ ls -l
total 12
drwxrwxr-x 3 piyush piyush 4096 Jan 20 21:44 dir1
-rw-rw-r-- 1 piyush piyush  12 Jan 20 21:54 file1.txt
-rw-rw-r-- 1 piyush piyush 424 Jan 20 23:04 output
piyush@CK-piyush:~/exercise$
```

12. Edit ~/.bashrc file such that when you change to "test" user it should clear the screen and print "Welcome".

Ans : First I will switch to test user then open the bash using nano command nano ~/.bashrc and then in the end I will append the text using echo "Welcome" .

So now whenever I use su - test first it asks the password and then displays the welcome message.

A terminal window with a dark background. The title bar is dark gray with a small icon on the left. The terminal content shows the word "Welcome" in white text on the first line. On the second line, the prompt "test@CK-piyush:~\$" is displayed in green text, followed by a white dollar sign "\$".

```
Welcome
test@CK-piyush:~$
```

13. Install "zip" package.

Ans : To install "zip" package I will use the sudo apt command . It will install the latest version automatically .

```
piyush@CK-piyush: ~/exercise
piyush@CK-piyush:~/exercise$ sudo apt install zip
[sudo] password for piyush:
Sorry, try again.
[sudo] password for piyush:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
zip is already the newest version (3.0-13ubuntu0.2).
zip set to manually installed.
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
piyush@CK-piyush:~/exercise$ zip --version
Copyright (c) 1990-2008 Info-ZIP - Type 'zip -L' for software license.
This is Zip 3.0 (July 5th 2008), by Info-ZIP.
Currently maintained by E. Gordon. Please send bug reports to
the authors using the web page at www.info-zip.org; see README for details.
Latest sources and executables are at ftp://ftp.info-zip.org/pub/infozip.
```

14. Compress "output" and "password_backup" files into a tar ball. List the files present inside the tar created.

Ans : To compress the files into a tar ball the command I use is `tar -czvf tar_files.tar.gz output /etc/passwd_backup`. I passed `/etc/passwd_backup` as the directory because the current directory didn't have the `password_backup` file. To list the files in the tarball the flag `-tvf` is passed and it lists all the files present inside the tarball.

```
piyush@CK-piyush: ~/exercise
piyush@CK-piyush:~/exercise$ tar -czvf tar_files.tar.gz output passwd_backup
output
tar: passwd_backup: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
piyush@CK-piyush:~/exercise$ tar -czvf tar_files.tar.gz output /etc/passwd_backup
output
tar: Removing leading '/' from member names
/etc/passwd_backup
tar: Removing leading '/' from hard link targets
piyush@CK-piyush:~/exercise$ tar -tvf tar_files.tar.gz
-rw----- piyush/piyush   436 2026-01-21 20:14 output
-rw-r--r-- root/root     2904 2026-01-20 22:15 etc/passwd_backup
piyush@CK-piyush:~/exercise$
```

15.scp this file to test user

Ans : The scp command can be used to copy the file to test user .

scp tar_files.tar test@localhost:/home/test/ because the test user is also present in the same machine so we do not need to get the remote address.

```
test@CK-piyush: ~  
piyush@CK-piyush:~/exercise$ scp tar_files.tar.gz test@localhost:/home/test/  
The authenticity of host 'localhost (127.0.0.1)' can't be established.  
ED25519 key fingerprint is SHA256:tdTXaJuHR9nhFo8rmXkh8f8JhJnDZBRqpPIE9YDAuk.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? y  
Please type 'yes', 'no' or the fingerprint: yes  
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.  
test@localhost's password:  
tar_files.tar.gz  
piyush@CK-piyush:~/exercise$ su - test  
Password:  
test@CK-piyush:~$ ls -al  
total 40  
drwxr-xr-x 5 test test 4096 Jan 21 21:33 .  
drwxr-xr-x 5 root root 4096 Jan 21 20:02 ..  
-rw-r--r-- 1 test test 428 Jan 21 21:06 .bash_history  
-rw-r--r-- 1 test test 220 Mar 31 2024 .bash_logout  
-rw-r--r-- 1 test test 3773 Jan 21 21:06 .bashrc  
drwxr-xr-x 2 test test 4096 Jan 21 21:33 .cache  
drwxrwxr-x 4 test test 4096 Jan 21 21:33 .local  
-rw-r--r-- 1 test test 807 Mar 31 2024 .profile  
drwxr-xr-x 3 test test 4096 Jan 21 21:33 .snap  
-rw-rw-r-- 1 test test 1227 Jan 21 21:33 tar_files.tar.gz  
test@CK-piyush:~$
```

16. Unzip this tar bar by logging into the remote server

Ans : Firstly I will login remotely using ssh command to the test server. Then using the ls command I will check that the tarball exist . Using tar with flags -xvf to unzip the tarball . This will do the job correctly.

```
test@CK-piyush:~$ ssh test@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:tDtXaJuHRR9nhFo8rmXkhBf8JhJnDZBRqpPIE9YDAuk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
test@localhost's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
7 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

test@CK-piyush:~$ ls -al
total 44
drwxr-x--- 6 test test 4096 Jan 21 21:46 .
drwxr-xr-x 5 root root 4096 Jan 21 20:02 ..
-rw----- 1 test test  428 Jan 21 21:06 .bash_history
-rw-r--r-- 1 test test  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 test test 3773 Jan 21 21:06 .bashrc
drwx----- 2 test test 4096 Jan 21 21:33 .cache
drwxrwxr-x 4 test test 4096 Jan 21 21:33 .local
-rw-r--r-- 1 test test  807 Mar 31  2024 .profile
drwx----- 3 test test 4096 Jan 21 21:33 snap
drwx----- 2 test test 4096 Jan 21 21:46 .ssh
-rw-rw-r-- 1 test test 1227 Jan 21 21:33 tar_files.tar.gz
test@CK-piyush:~$ tar -xvf tar_files.tar.gz
output
etc/passwd_backup
test@CK-piyush:~$ ls -al
total 52
drwxr-x--- 7 test test 4096 Jan 21 21:46 .
drwxr-xr-x 5 root root 4096 Jan 21 20:02 ..
-rw----- 1 test test  428 Jan 21 21:06 .bash_history
-rw-r--r-- 1 test test  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 test test 3773 Jan 21 21:06 .bashrc
drwx----- 2 test test 4096 Jan 21 21:33 .cache
drwxrwxr-x 4 test test 4096 Jan 21 21:33 .local
-rw-r--r-- 1 test test  807 Mar 31  2024 .profile
drwx----- 3 test test 4096 Jan 21 21:33 snap
drwx----- 2 test test 4096 Jan 21 21:46 .ssh
-rw-rw-r-- 1 test test 1227 Jan 21 21:33 tar_files.tar.gz
```

17. Download any image from web and move to desktop

Ans: To download image from the web the command used is `wget` followed by the filename to be downloaded. The file is downloaded in the exercise directory as this is the current directory i am in , so next step is to move the image in the Desktop directory so I used `mv` command to move the image to Desktop.

```
piyush@CK-piyush: ~/Desktop
piyush@CK-piyush: ~/exercise$ wget https://t4.ftcdn.net/jpg/03/89/39/09/360_F_389390965_lwqVX10TBdFH2WMBH6GGF8pcNiOhzfmb.webp
--2026-01-21 21:59:44-- https://t4.ftcdn.net/jpg/03/89/39/09/360_F_389390965_lwqVX10TBdFH2WMBH6GGF8pcNiOhzfmb.webp
Resolving t4.ftcdn.net (t4.ftcdn.net)... 199.232.21.91
Connecting to t4.ftcdn.net (t4.ftcdn.net)|199.232.21.91|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9878 (9.6K) [image/webp]
Saving to: '360_F_389390965_lwqVX10TBdFH2WMBH6GGF8pcNiOhzfmb.webp'

360_F_389390965_lwqVX10TBdFH2WMBH6GGF8pcNiOhz 100%[=====] 9.65K ---KB/s in 0.02s

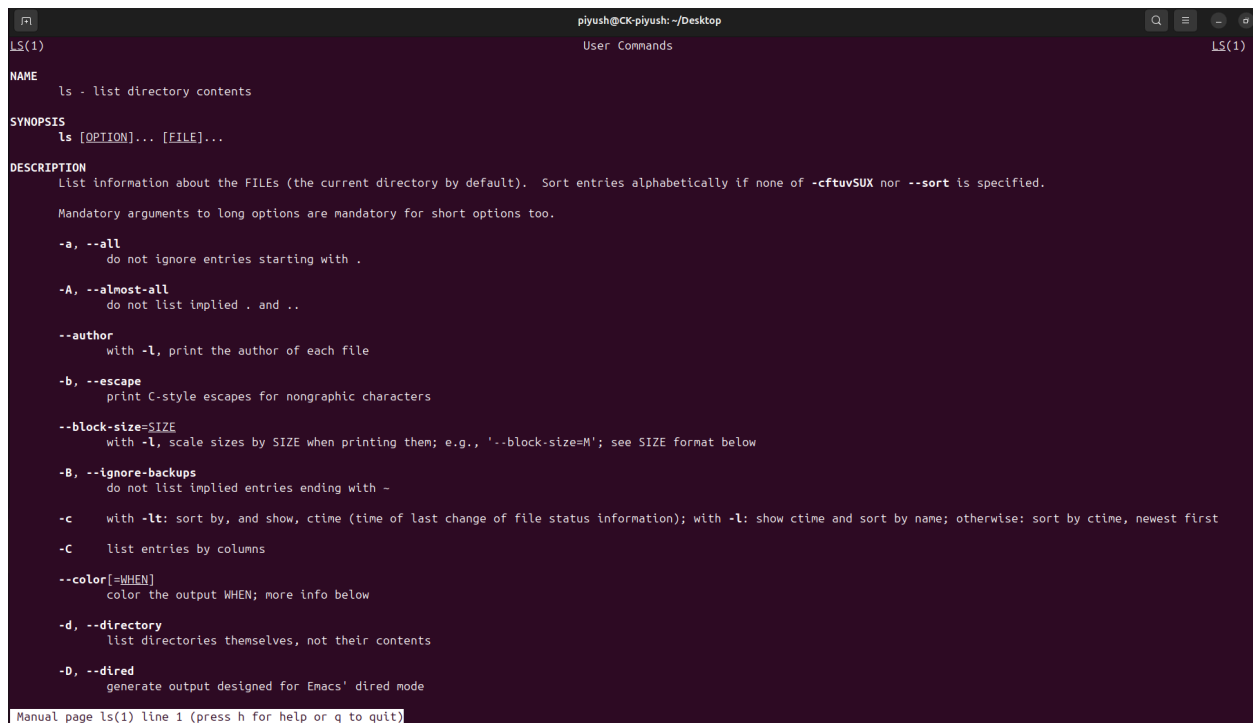
2026-01-21 21:59:44 (506 KB/s) - '360_F_389390965_lwqVX10TBdFH2WMBH6GGF8pcNiOhzfmb.webp' saved [9878/9878]

piyush@CK-piyush: ~/exercise$ ls -al
total 36
drwxrwxr-x 3 piyush piyush 4096 Jan 21 21:59 .
drwxr-x--- 16 piyush piyush 4096 Jan 21 21:55 ..
-rw-rw-r-- 1 piyush piyush 9878 Nov 1 2020 360_F_389390965_lwqVX10TBdFH2WMBH6GGF8pcNiOhzfmb.webp
drwxrwxr-x 3 piyush piyush 4096 Jan 20 21:44 d1c1
-rw-rw-r-- 1 piyush piyush 12 Jan 20 21:54 file1.txt
-rw-rw-r-- 1 piyush piyush 436 Jan 21 20:14 output
-rw-rw-r-- 1 piyush piyush 1227 Jan 21 21:17 tar_files.tar.gz
piyush@CK-piyush: ~/exercise$ mv 360_F_389390965_lwqVX10TBdFH2WMBH6GGF8pcNiOhzfmb.webp /home/piyush/Desktop
piyush@CK-piyush: ~/exercise$ cd ..
piyush@CK-piyush: ~/Desktop$ cd Desktop
piyush@CK-piyush: ~/Desktop$ ls -al
total 24
drwxr-xr-x 3 piyush piyush 4096 Jan 21 22:00 .
drwxr-x--- 16 piyush piyush 4096 Jan 21 21:55 ..
-rw-rw-r-- 1 piyush piyush 9878 Nov 1 2020 360_F_389390965_lwqVX10TBdFH2WMBH6GGF8pcNiOhzfmb.webp
drwxrwxr-x 2 piyush piyush 4096 Jan 21 14:30 Assignments
piyush@CK-piyush: ~/Desktop$
```

18.How to get help of commands usages.

Ans : To get help of command usages the linux provides man command to see the manual page for the said command . To use this we simply do man followed by command name and this open up the manual page of the command passed to get usage details and flags of the command .

Like in the following screenshot attached I use man ls command to get usage regarding ls command and it display the page that shows the details about the ls command and the options to use with it.



```
piyush@CK-piyush: ~/Desktop
LS(1) User Commands LS(1)

NAME
  ls - list directory contents

SYNOPSIS
  ls [OPTION]... [FILE]...

DESCRIPTION
  List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

  Mandatory arguments to long options are mandatory for short options too.

  -a, --all
    do not ignore entries starting with .

  -A, --almost-all
    do not list implied . and ..

  --author
    with -l, print the author of each file

  -b, --escape
    print C-style escapes for nongraphic characters

  --block-size=SIZE
    with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

  -B, --ignore-backups
    do not list implied entries ending with ~

  -c      with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first

  -C      list entries by columns

  --color[=WHEN]
    color the output WHEN; more info below

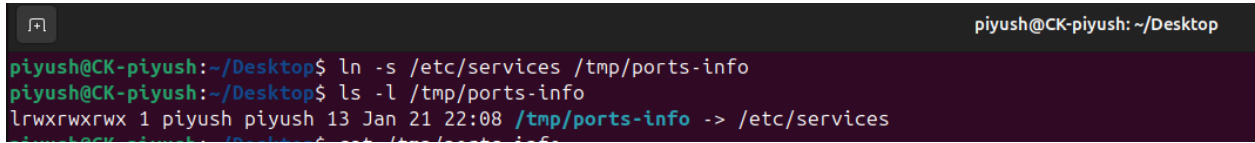
  -d, --directory
    list directories themselves, not their contents

  -D, --dired
    generate output designed for Emacs' dired mode

Manual page ls(1) line 1 (press h for help or q to quit)
```


19. Create a symlink of /etc/services into /tmp/ports-info

Ans : To create the symlink , the command used is ln with flag -s to create the link i.e. it creates a link between /tmp/ports-info pointing to the original file /etc/services.

A terminal window with a dark background. The title bar shows a window icon and the text "piyush@CK-piyush: ~/Desktop". The terminal contains three lines of text: a command to create a symlink, a command to list the file with long options, and the output of that command.

```
piyush@CK-piyush:~/Desktop$ ln -s /etc/services /tmp/ports-info
piyush@CK-piyush:~/Desktop$ ls -l /tmp/ports-info
lrwxrwxrwx 1 piyush piyush 13 Jan 21 22:08 /tmp/ports-info -> /etc/services
```

To verify the successful creation of the link , I use ls -l and it shows that the files are linked .

20. You are appointed as a Software Engineer in ABC media services. On your first day you need to troubleshoot a problem. There is a command "xyz" somewhere installed in that linux system. But as a new joiner you do not have any idea about where it is installed. How can you check that?

Ans : Two commands come in handy for doing the task, 'which' and 'whereis' command. The only difference between these two commands is that 'whereis' command will search for executable files and manual pages using system built database whereas 'which' command does this by searching the path for the executable files.

```
piyush@CK-piyush: ~/Desktop
piyush@CK-piyush:~/Desktop$ # which xyz
piyush@CK-piyush:~/Desktop$ # whereis xyz
piyush@CK-piyush:~/Desktop$ where is /etc
where: command not found
piyush@CK-piyush:~/Desktop$ whereis /etc
etc: /usr/local/etc
piyush@CK-piyush:~/Desktop$ which /etc
piyush@CK-piyush:~/Desktop$ which ls
/usr/bin/ls
piyush@CK-piyush:~/Desktop$
```